Project: Worked examples
Title: Glaser condensation (interstitial and surface)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Interstitial condensation in a layered wall by the Glaser method over twelve monthly climate steps: the temperature and saturation pressure at every layer interface, the vapour-pressure line with its tangent construction at the condensation planes, the condensate on each interface month by month carried forward from the first month of condensation (drying when the weather turns), the annual balance, the peak accumulation against the owner's limit, and the surface temperature factor f_Rsi against its minimum at the critical surface humidity.
Layer resistance and diffusion thickness: R_j = d_j/λ_j; s_d,j = μ_j·d_j Total resistance and interface temperature: R_T = R_si + Σ R_j + R_se; θ_k = θ_i − (θ_i − θ_e)·(R_si + Σ_(j≤k) R_j)/R_T Saturation pressure (Magnus): p_sat = 610.5·exp(17.269·θ/(237.3 + θ)) over water (θ ≥ 0 °C); p_sat = 610.5·exp(21.875·θ/(265.5 + θ)) over ice (θ < 0 °C) Outside vapour pressure: p_e = RH_e·p_sat(θ_e) Inside vapour pressure: p_i = RH_i·p_sat(θ_i), or p_e + f_Δp·Δv(θ_e)·R_v·(T_i + T_e)/2 with the moisture excess Δv linear between two user points and constant beyond them (R_v = 461.5 J/(kg·K)) and the safety factor f_Δp on the pressure difference (a record, default 1.10; clause wording to be checked against the owner's copy of ISO 13788) Vapour flux through a layer: g = δ0·Δp/Δs_d with δ0 = 2×10^{-10} kg/(m·s·Pa) Tangent construction: the line from p_i to p_e is bent onto the saturation line at the interfaces it would rise above (and at wet interfaces), by the smallest-slope rule from each contact point Condensation at an interface: g_c = g_in − g_out; Mc accumulates g_c·t over 12 months from the first month of condensation (t = the row's days, else the calendar days of the month, January to December order), a wet interface (held at saturation) drying at g_c < 0 until nothing is left Annual balance and peak: PASS when no condensate remains after the 12 months; Mc_max = the most held on any interface, checked against the user's limit Surface factor: f_Rsi = 1 − R_si,f/R_T,f with R_si,f = 0.25 m²K/W by default for an opaque element (conventional, a record; 0.13 for glazing); f_Rsi,min = the largest monthly (θ_si,min − θ_e)/(θ_i − θ_e) with p_sat(θ_si,min) = p_i/φ_crit; PASS when f_Rsi ≥ f_Rsi,min

Ref: ISO 13788 cl. 5 (clause number to be checked against the owner's copy) — Hygrothermal performance of building components and elements — internal surface temperature and surface condensation
Ref: ISO 13788 cl. 6 (clause number to be checked against the owner's copy) — Hygrothermal performance of building components and elements — interstitial condensation (Glaser method)
Ref: ISO 13788 moisture excess and its design margin on Δp (clause number and wording to be checked against the owner's copy) — Hygrothermal performance of building components and elements — internal humidity from the humidity class
Ref: EN 15026 whole standard (named as the recommended method for timber and breathable builds) — Hygrothermal performance of building components — assessment of moisture transfer by numerical simulation

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
layers	Wall layers from the inside outwards (d, λ, μ; the shape of the B4 layers table plus μ)	3 row(s)		drawing: A-310 rev B	any
layers[1]		name = Plasterboard; d = 12.5 mm; lambda = 0.25 W/(m*K); mu = 8			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
layers[2]		name = Mineral wool; d = 50 mm; lambda = 0.035 W/(m*K); mu = 1			
layers[3]		name = Existing brick; d = 215 mm; lambda = 0.8 W/(m*K); mu = 10			
Rsi	Internal surface resistance R_si	0.13	m^2*K/W	spec: Project surface-resistance record (internal, horizontal flow), assumed	0.04 … 0.25 m^2*K/W
Rse	External surface resistance R_se	0.04	m^2*K/W	spec: Project surface-resistance record (external), assumed	0.02 … 0.25 m^2*K/W
Rsi_f	Internal surface resistance R_si,f for the f_Rsi check (0.25 m²K/W is the conventional value for an opaque element, a record to check against the owner's copy of ISO 13788; 0.13 for glazing)	0.25	m^2*K/W	spec: Conventional R_si,f = 0.25 m²K/W for the surface-condensation check of an opaque element (a record, to be checked against the owner's copy of ISO 13788), assumed	0.04 … 0.5 m^2*K/W
theta_int	Internal air temperature	20	degC	spec: SPEC-A-02 §3.1: living-room design temperature	10 … 30 degC
humidity_mode	Internal humidity from	constant internal relative humidity		assumption: Constant internal humidity; no humidity-class record is held for this project.	any
RH_int	Internal relative humidity (constant-humidity mode)	50	%	spec: SPEC-A-02 §3.1: internal design humidity	20 … 90 %

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
dp_f	Safety factor on the vapour-pressure difference Δp = Δv·R_v·(T_i + T_e)/2 in the moisture-excess mode (design margin, 1.10 by convention; a record, wording to be checked against the owner's copy of ISO 13788)	1.1		default	1 … 1.5
climate	Twelve monthly outside conditions (the user's climate record, in the order the year is run)	12 row(s)		assumption: Invented illustrative maritime year; replace with the project's own monthly climate record (take it from your licensed source).	any
climate[1]		month = M1; theta_e = 2 degC; RH_e = 90 %			
climate[2]		month = M2; theta_e = 3 degC; RH_e = 88 %			
climate[3]		month = M3; theta_e = 6 degC; RH_e = 84 %			
climate[4]		month = M4; theta_e = 9 degC; RH_e = 80 %			
climate[5]		month = M5; theta_e = 13 degC; RH_e = 76 %			

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
climate[6]		month = M6; theta_e = 16 degC; RH_e = 74 %			
climate[7]		month = M7; theta_e = 18 degC; RH_e = 74 %			
climate[8]		month = M8; theta_e = 18 degC; RH_e = 76 %			
climate[9]		month = M9; theta_e = 15 degC; RH_e = 82 %			
climate[10]		month = M10; theta_e = 11 degC; RH_e = 87 %			
climate[11]		month = M11; theta_e = 6 degC; RH_e = 90 %			
climate[12]		month = M12; theta_e = 3 degC; RH_e = 91 %			
RH_crit	Critical surface relative humidity	80	%	spec: SPEC-A-02 §4.4: critical surface humidity	50 … 100 %

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
Mc_limit	Limit on the condensate on any interface (blank = report only)	500	g/m^2	spec: SPEC-A-02 §4.4: limit on the condensate per interface	1 ... 5000 g/m^2
breathable	The build-up is timber or breathable (hygroscopic layers, capillary transport)	no		assumption: Mineral wool and brick: no timber or hygroscopic layers.	any

3. Assumptions

A1. Steady state in each month: one-dimensional diffusion of water vapour through homogeneous layers in series, with the monthly mean climate; no air movement, no liquid transport, no moisture storage, no solar or rain gain. (The Glaser method.)
A2. Material properties (λ, μ) are constants of the layer, not functions of its moisture or temperature; condensate does not change them. (The Glaser method.)
A3. Condensation is checked at the interfaces between layers only; a thick layer where the plane would lie inside it should be split into thinner layers. The saturation line is joined straight between the interface values. (Stated limit.)
A4. The Magnus saturation pressure is used over water at and above 0 °C and over ice below, and the water-vapour permeability of air is δ0 = 2×10^{-10} kg/(m·s·Pa); both are closed-form physics, not data of a standard. (Derived.)
A5. Where a month's length is blank it is the calendar days of the month, taking the climate rows as January to December (February 28 days); enter the days where the record starts in another month. A wet interface that dries within a month is taken as dry from the next month; the month's rates are those of the wet state at its start. (Monthly steps.)
A6. The monthly climate, the humidity-class points, the surface resistances, the critical humidity and the limit on the condensate are the user's records from the user's own licensed copies; no climate data, humidity-class table, material table or limit of any standard is shipped. (No table from a standard is shipped.)
A7. humidity_mode (Internal humidity from) is taken as constant internal relative humidity. (Constant internal humidity; no humidity-class record is held for this project.)
A8. climate (Twelve monthly outside conditions (the user's climate record, in the order the year is run)) is taken as 12 row(s). (Invented illustrative maritime year; replace with the project's own monthly climate record (take it from your licensed source).)
A9. breathable (The build-up is timber or breathable (hygroscopic layers, capillary transport)) is taken as no. (Mineral wool and brick: no timber or hygroscopic layers.)
A10. Month length: 12 row(s) of the climate gave no days, so the calendar days of the month are used with the rows taken as January to December (February 28 days). If the record starts in another month, enter the days of each row. (Calendar months.)

4. Working

[8] R_1 = d / □ ISO 13788 cl. 6
= Plasterboard: 12.5 mm / 0.25 W/(m·K)
= 0.05 m²·K/W
Plasterboard: thermal resistance

[9] sd_1 = □ · d ISO 13788 cl. 6
= Plasterboard: 8 × 12.5 mm
= 0.1 m
Plasterboard: diffusion-equivalent air layer thickness

[10] R_2 = d / λ ISO 13788 cl. 6
= Mineral wool: 50 mm / 0.035 W/(m·K)
= 1.4286 m²·K/W
Mineral wool: thermal resistance
[11] sd_2 = μ · d ISO 13788 cl. 6
= Mineral wool: 1 × 50 mm
= 0.05 m
Mineral wool: diffusion-equivalent air layer thickness
[12] R_3 = d / λ ISO 13788 cl. 6
= Existing brick: 215 mm / 0.8 W/(m·K)
= 0.26875 m²·K/W
Existing brick: thermal resistance
[13] sd_3 = μ · d ISO 13788 cl. 6
= Existing brick: 10 × 215 mm
= 2.15 m
Existing brick: diffusion-equivalent air layer thickness
[14] R_T = R_si + Σ R_j + R_se ISO 13788 cl. 6
= 0.13 + 1.7473 + 0.04
= 1.9173 m²·K/W
Total thermal resistance
[15] S_d = Σ s_d,j ISO 13788 cl. 6
= 2.3 m
= 2.3 m
Total diffusion-equivalent air layer thickness
[16] psat_i = 610.5·exp(17.269·ϑ/(237.3 + ϑ)) ISO 13788 cl. 6
= ϑ = 20 °C
= 2337 Pa
Saturation vapour pressure at the internal air temperature (Magnus)
[17] p_ext_m = RH_e · p_sat(ϑ_e) ISO 13788 cl. 6
= months 1 … 12 of the climate
= [634.76, 666.5, 785.09, 917.96, 1137.7, 1344.8, 1526.5, 1567.8, 1397.6, 1141.4, 841.16, 689
Outside vapour pressure by month
[18] p_int_m = RH_i · p_sat(ϑ_i) ISO 13788 cl. 6
= 50 % × 2337 Pa
= [1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 11
Inside vapour pressure by month
[19] n_start = first month of condensation ISO 13788 cl. 6
= M11 (row 11)
= 11
Row of the climate in which the accumulation starts (0 when nothing condenses)
[20] Mc_1 = Σ Mc over the interfaces ISO 13788 cl. 6
= M1: ϑ_e = 2 °C; 1023.7 g/m² gained (-: dried) this month
= 2069.8 g/m²
M1: condensate on the wall at the end of the month
[21] Mc_2 = Σ Mc over the interfaces ISO 13788 cl. 6
= M2: ϑ_e = 3 °C; 752.09 g/m² gained (-: dried) this month
= 2821.9 g/m²
M2: condensate on the wall at the end of the month
[22] Mc_3 = Σ Mc over the interfaces ISO 13788 cl. 6
= M3: ϑ_e = 6 °C; 200.73 g/m² gained (-: dried) this month
= 3022.6 g/m²
M3: condensate on the wall at the end of the month

[23] Mc_4 = ▯ Mc over the interfaces ISO 13788 cl. 6
= M4: ▯_e = 9 °C; -517.84 g/m² gained (-: dried) this month
= 2504.8 g/m²
M4: condensate on the wall at the end of the month
[24] Mc_5 = ▯ Mc over the interfaces ISO 13788 cl. 6
= M5: ▯_e = 13 °C; -1698.2 g/m² gained (-: dried) this month
= 806.63 g/m²
M5: condensate on the wall at the end of the month
[25] Mc_6 = ▯ Mc over the interfaces ISO 13788 cl. 6
= M6: ▯_e = 16 °C; -806.63 g/m² gained (-: dried) this month
= 0 g/m²
M6: condensate on the wall at the end of the month
[26] Mc_7 = ▯ Mc over the interfaces ISO 13788 cl. 6
= M7: ▯_e = 18 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M7: condensate on the wall at the end of the month
[27] Mc_8 = ▯ Mc over the interfaces ISO 13788 cl. 6
= M8: ▯_e = 18 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M8: condensate on the wall at the end of the month
[28] Mc_9 = ▯ Mc over the interfaces ISO 13788 cl. 6
= M9: ▯_e = 15 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M9: condensate on the wall at the end of the month
[29] Mc_10 = ▯ Mc over the interfaces ISO 13788 cl. 6
= M10: ▯_e = 11 °C; 0 g/m² gained (-: dried) this month
= 0 g/m²
M10: condensate on the wall at the end of the month
[30] Mc_11 = ▯ Mc over the interfaces ISO 13788 cl. 6
= M11: ▯_e = 6 °C; 207.78 g/m² gained (-: dried) this month
= 207.78 g/m²
M11: condensate on the wall at the end of the month
[31] Mc_12 = ▯ Mc over the interfaces ISO 13788 cl. 6
= M12: ▯_e = 3 °C; 838.33 g/m² gained (-: dried) this month
= 1046.1 g/m²
M12: condensate on the wall at the end of the month
[32] Mc_if_1 = max over the year of Mc ISO 13788 cl. 6
= between 'Plasterboard' and 'Mineral wool'
= 0 g/m²
Most condensate held on the interface between 'Plasterboard' and 'Mineral wool'
[33] Mc_if_2 = max over the year of Mc ISO 13788 cl. 6
= between 'Mineral wool' and 'Existing brick'
= 3022.6 g/m²
Most condensate held on the interface between 'Mineral wool' and 'Existing brick'
[34] Mc_max = max over interfaces and months of Mc ISO 13788 cl. 6
= 3022.6 g/m²
= 3022.6 g/m²
Most condensate held on any one interface in the year
[35] Mc_tot = max over months of ▯ Mc ISO 13788 cl. 6
= 3022.6 g/m²
= 3022.6 g/m²
Most condensate on the whole wall in the year

[36] Mc_end = □ Mc after 12 months — ISO 13788 cl. 6
= 0 g/m²
= 0 g/m²
Condensate left on the wall at the end of the 12-month cycle
[37] n_diag = month drawn
= M3 (row 3)
= 3
Row of the climate the diagram vectors are for
[38] sd_if = □ s_d to the node — ISO 13788 cl. 6
= 4 nodes, inside surface first
= [0, 0.1, 0.15, 2.3] m
Diffusion thickness at the surfaces and interfaces
[39] x_if = □ d to the node — ISO 13788 cl. 6
= 4 nodes
= [0, 0.0125, 0.0625, 0.2775] m
Depth from the inside surface at the surfaces and interfaces
[40] theta_if = □_i - (□_i - □_e)·(R_si + □ R)/R_T — ISO 13788 cl. 6
= □_e = 6 °C
= [19.051, 18.686, 8.2544, 6.2921] degC
Temperature at the surfaces and interfaces
[41] psat_if = 610.5·exp(17.269·□/(237.3 + □)), ice below 0 °C — ISO 13788 cl. 6
= 4 nodes
= [2203.1, 2153.5, 1090.9, 953.69] Pa
Saturation vapour pressure at the surfaces and interfaces (the Glaser saturation line)
[42] p_if = tangent construction — ISO 13788 cl. 6
= 4 nodes
= [1168.5, 1116.8, 1090.9, 785.09] Pa
Vapour pressure at the surfaces and interfaces after the tangent construction
[43] R_Tf = R_si,f + □ R_j + R_se — ISO 13788 cl. 5
= 0.25 + 1.7473 + 0.04
= 2.0373 m²·K/W
Total thermal resistance with the internal surface resistance of the surface check
[44] fRsi = 1 - R_si/R_T — ISO 13788 cl. 5
= 1 - 0.25 / 2.0373
= 0.87729
Temperature factor of the internal surface
[45] fRsi_min = max over months of (□_si,min - □_e)/(□_i - □_e) — ISO 13788 cl. 5
= □_si,min from p_i/□_crit; governing month: M1
= 0.59026
Minimum temperature factor at the critical surface humidity
[46] n_fcrit = month that governs f_Rsi,min — ISO 13788 cl. 5
= M1 (row 1)
= 1
Row of the climate that governs the surface check (0 when none applies)
[48] u_Mc = Mc_max / Mc_limit
= 3022.6 g/m² / 500 g/m²
= 6.0452
Utilisation of the limit on the condensate per interface
[49] u_f = f_Rsi,min / f_Rsi — ISO 13788 cl. 5
= 0.59026 / 0.87729
= 0.67282
Utilisation of the surface temperature factor

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[50] pass = every check passes
= 3 check(s)
= 0
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1 when the annual balance, the limit and the surface factor all pass, else 0

5. Results

SYMBOL	VALUE	DESCRIPTION
R_T	1.9173 m²·K/W	Total thermal resistance
S_d	2.3 m	Total diffusion-equivalent air layer thickness
psat_i	2337 Pa	Saturation vapour pressure at the internal temperature
p_ext_m	[634.76, 666.5, 785.09, 917.96, 1137.7, 1344.8, 1526.5, 1567.8, 1397.6, 1141.4, 841.16, 689.22] Pa	Outside vapour pressure by month
p_int_m	[1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5, 1168.5] Pa	Inside vapour pressure by month
n_start	11	First month of condensation
Mc_1	2069.8 g/m²	M1: condensate on the wall
Mc_2	2821.9 g/m²	M2: condensate on the wall
Mc_3	3022.6 g/m²	M3: condensate on the wall
Mc_4	2504.8 g/m²	M4: condensate on the wall
Mc_5	806.63 g/m²	M5: condensate on the wall
Mc_6	0 g/m²	M6: condensate on the wall
Mc_7	0 g/m²	M7: condensate on the wall
Mc_8	0 g/m²	M8: condensate on the wall
Mc_9	0 g/m²	M9: condensate on the wall
Mc_10	0 g/m²	M10: condensate on the wall
Mc_11	207.78 g/m²	M11: condensate on the wall
Mc_12	1046.1 g/m²	M12: condensate on the wall
Mc_if_1	0 g/m²	Peak condensate on interface 1
Mc_if_2	3022.6 g/m²	Peak condensate on interface 2
Mc_max	3022.6 g/m²	Most condensate on any interface
Mc_tot	3022.6 g/m²	Most condensate on the wall
Mc_end	0 g/m²	Condensate left after 12 months
n_diag	3	Month of the diagram
sd_if	[0, 0.1, 0.15, 2.3] m	Diffusion thickness at the nodes
x_if	[0, 0.0125, 0.0625, 0.2775] m	Depth at the nodes

SYMBOL	VALUE	DESCRIPTION
theta_if	[19.051, 18.686, 8.2544, 6.2921] degC	Temperature at the nodes
psat_if	[2203.1, 2153.5, 1090.9, 953.69] Pa	Saturation pressure at the nodes
p_if	[1168.5, 1116.8, 1090.9, 785.09] Pa	Vapour pressure at the nodes
fRsi	0.87729	Temperature factor of the internal surface
fRsi_min	0.59026	Minimum temperature factor
n_fcrit	1	Month governing the surface check
R_Tf	2.0373 m²·K/W	Total resistance for the surface check
u_Mc	6.0452	Peak condensate against its limit
u_f	0.67282	Surface factor against its minimum
pass	0	All checks pass

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Peak condensate on an interface against its limit (500 g/m²)	u_Mc = 6.0452	u_Mc ≤ 1	6.05	ISO 13788 cl. 6
PASS	Surface condensation: f_Rsi against f_Rsi,min at 80 % critical humidity	u_f = 0.67282	u_f ≤ 1	0.673	ISO 13788 cl. 5
PASS	Annual moisture balance: the condensate dries out within the year	Mc_end = 0 g/m²	Mc_end ≤ 0 g/m²	0	ISO 13788 cl. 6

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: layers (Layer conductivity and vapour resistance); Rsi, Rse, Rsi_f (Surface resistances); climate (Monthly outside climate); RH_crit, Mc_limit (Condensation criteria). Enter each as one so its source, edition and licence travel with it.
WARNING — input.dp_f: dp_f (Safety factor on the vapour-pressure difference Δp = Δv·R_v·(T_i + T_e)/2 in the moisture-excess mode (design margin, 1.10 by convention; a record, wording to be checked against the owner's copy of ISO 13788)) was not entered; the template default 1.1 is used.

8. Verification

Template bs.glaser@1.1.0 Content hash fef5be08ad1b9421 Engine calc-engine/1.0.0
Known-answer cases passed 7/7 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0